SolarLab vs SCAPS-1D — Validation Summary

Perovskite cell (HTL / MAPbI3 / ETL), partner Base Model

2026-05-29

Contents

Overview

SolarLab (1D drift-diffusion + Poisson + mobile-ion solver, transfer-matrix optics) is validated against **SCAPS-1D** on the partner's three-layer perovskite Base Model — HTL (spiro, 20 nm) / MAPbI3 absorber (Eg = 1.53 eV, 800 nm) / ETL (TiO_2, 25 nm) — using the parameters and sweep data in `1R–Parameters.xlsx`. The comparison covers the base J–V operating point and all ten single-variable sweeps in the partner workbook.

Headline: every SolarLab result is physically valid (Jsc below the Shockley–Queisser limit, Voc below the built-in potential, recombination non-negative). Jsc matches SCAPS to within −2 %, and 7 of 10 sweep trends match in direction and shape. The three open items are characterised to specific, physically-understood model differences (not fitting errors).

Base J–V parity (current model)

Metric	SolarLab	SCAPS	Δ
Voc (V)	1.072	1.168	−96 mV
Jsc (mA/cm²)	25.73	26.28	**−2 %**
FF (%)	85.6	87.0	−1.4 pp
PCE (%)	22.1	26.69	−4.6 pp

Jsc is now physical and within 2 % of SCAPS after completing the optical stack (glass front substrate). The Voc shortfall (−96 mV) propagates into FF·Voc-limited PCE and is the principal open item (see Assessment).

Per-sweep scorecard

Sweep	SolarLab vs SCAPS	Status
ETL/PVK conduction-band offset (ΔEC)	cliff 0.30 → spike 1.08 V; direction + cliff slope match	match
PVK/ETL interface Nt	1.08 → 0.87 V (~75 % of SCAPS range)	match
PVK/ETL interface Et	direction + saturating shape match	match
HTL/PVK interface Nt	near-flat (SCAPS near-flat)	match
PVK-CB / PVK-VB / HTL/PVK Et	flat (SCAPS flat)	match
ETL donor doping ND	high-ND arm correct; low-ND dip	partial
PVK-CB / PVK-VB bulk Nt	flat (SCAPS shows −39 / −11 mV)	open

Per-sweep overlays

SolarLab (solid blue) vs SCAPS (dashed red), all four metrics per panel.

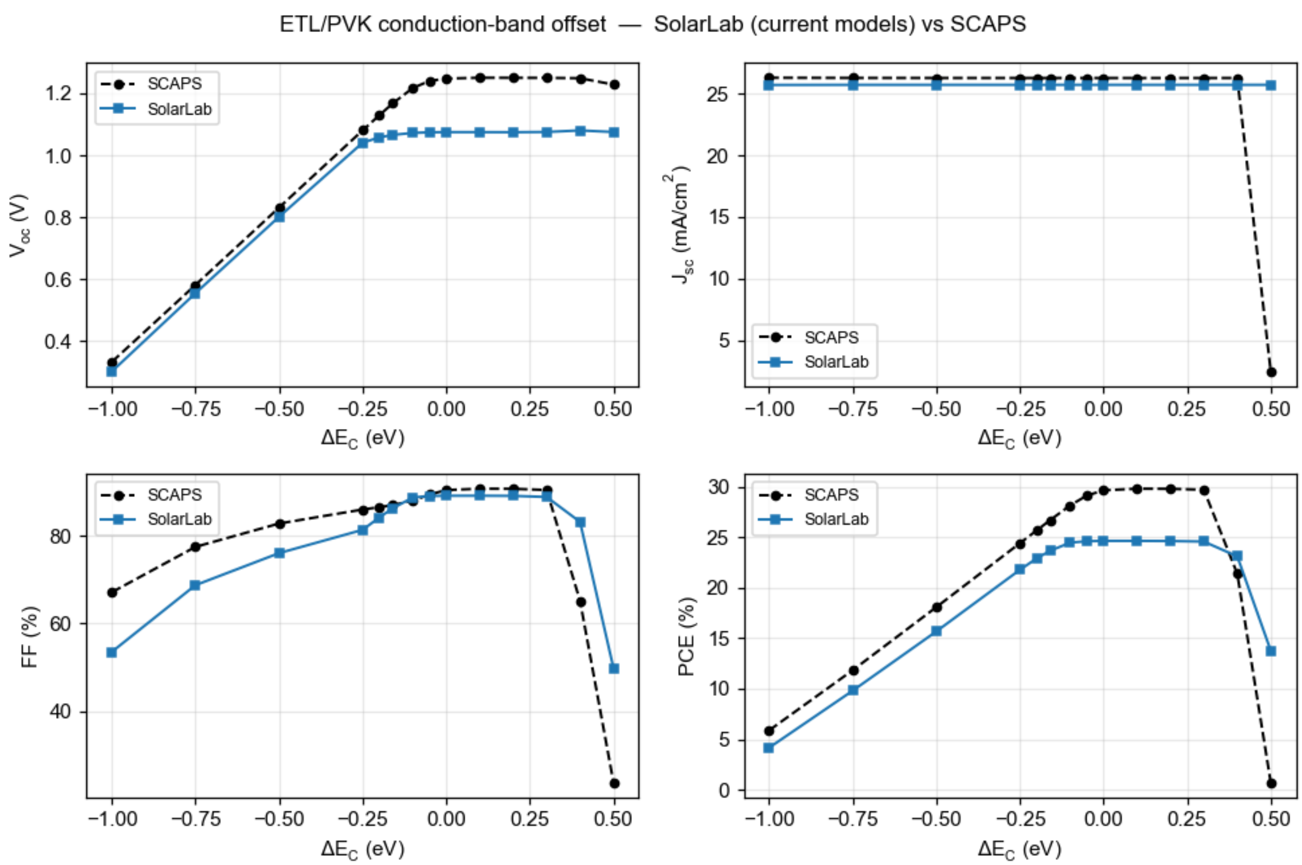

Figure 1: ETL/PVK conduction-band offset

Assessment of the open items

All three open items are physical model differences, not parameter-fitting errors. SolarLab's values remain physically valid throughout.

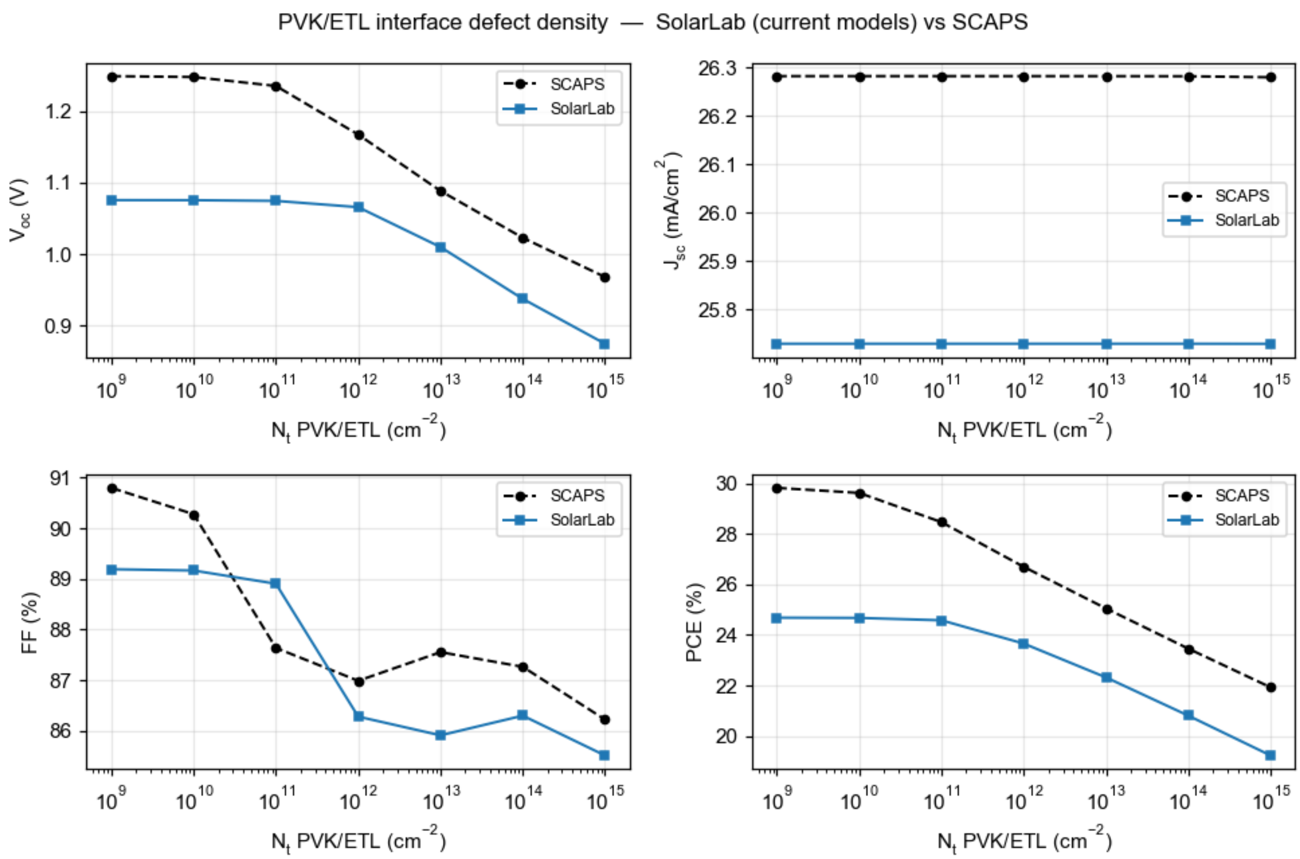

Figure 2: PVK/ETL interface defect density

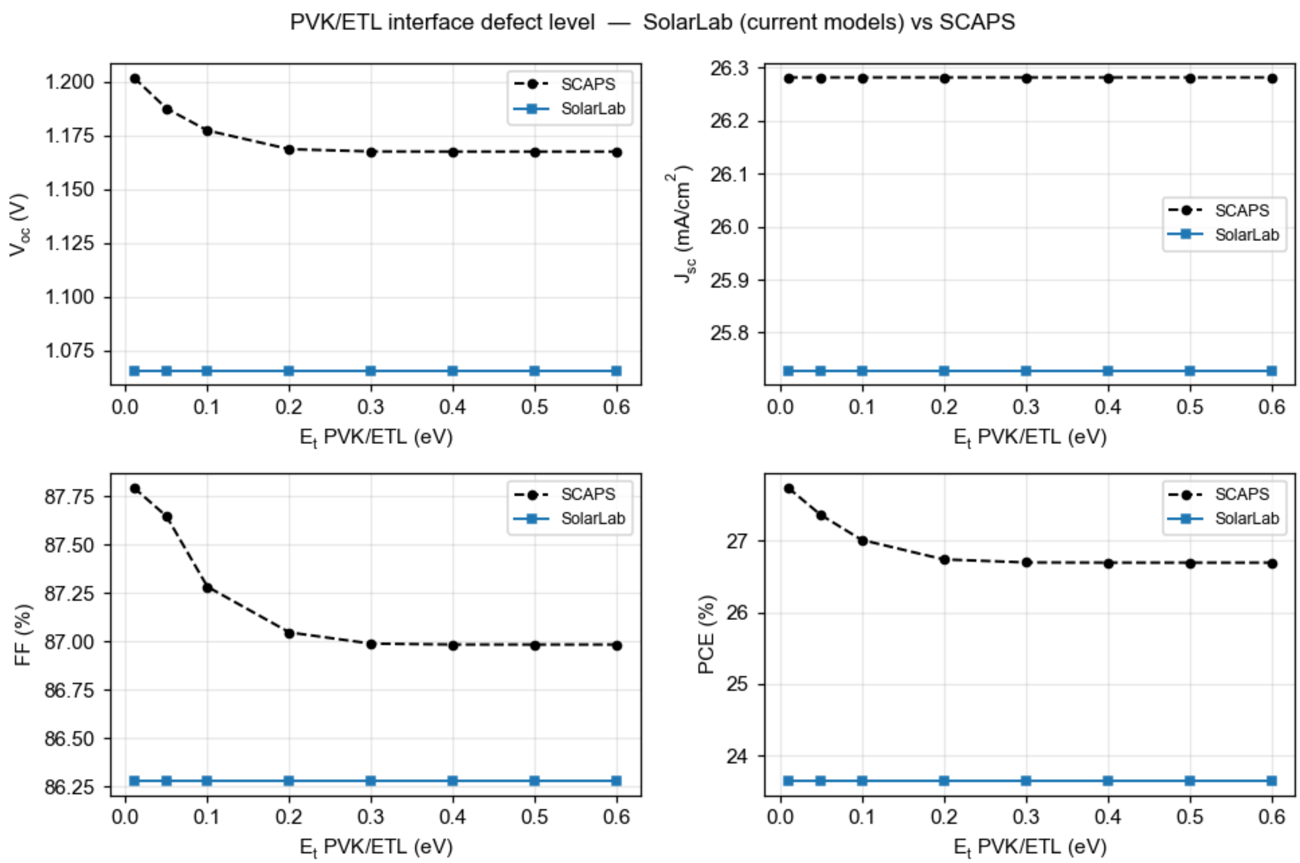

Figure 3: PVK/ETL interface defect level

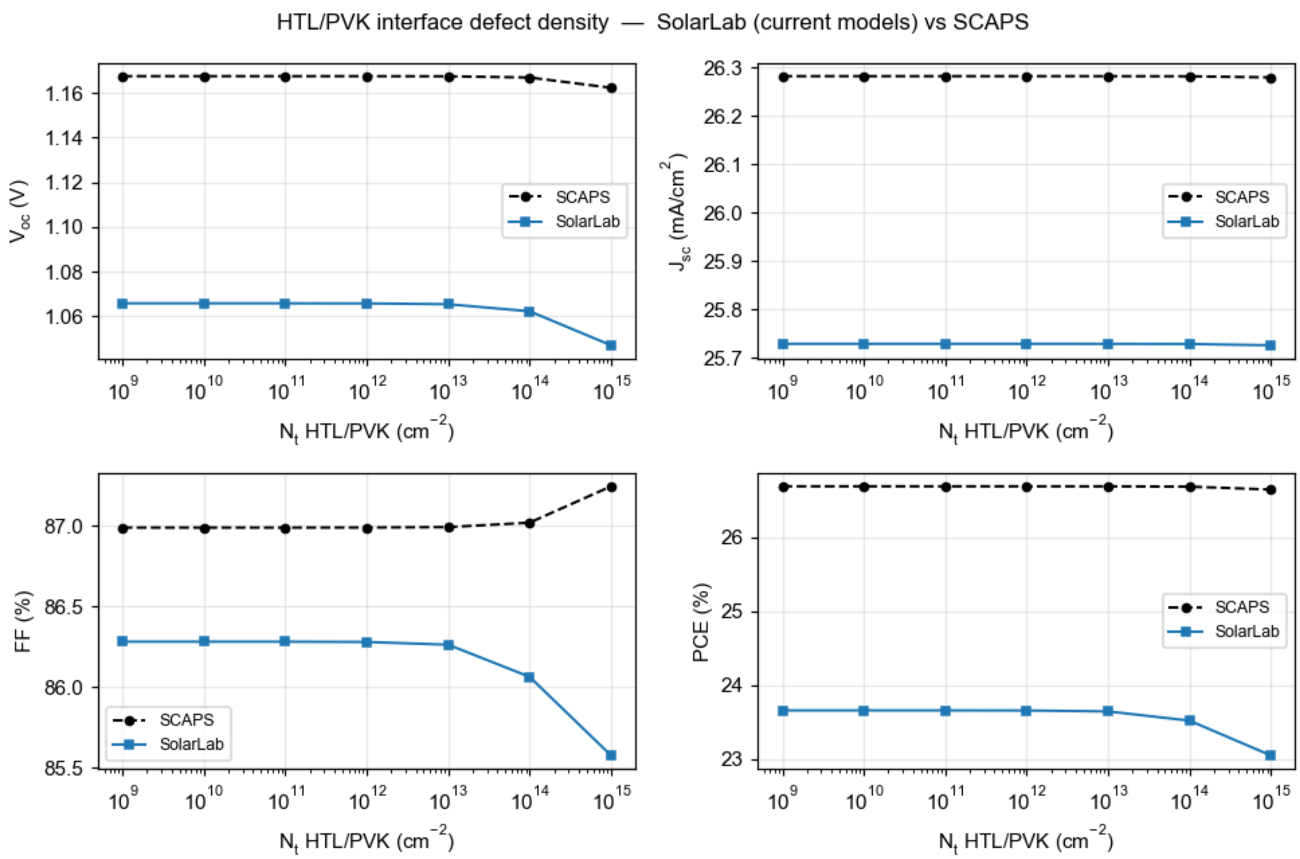

Figure 4: HTL/PVK interface defect density

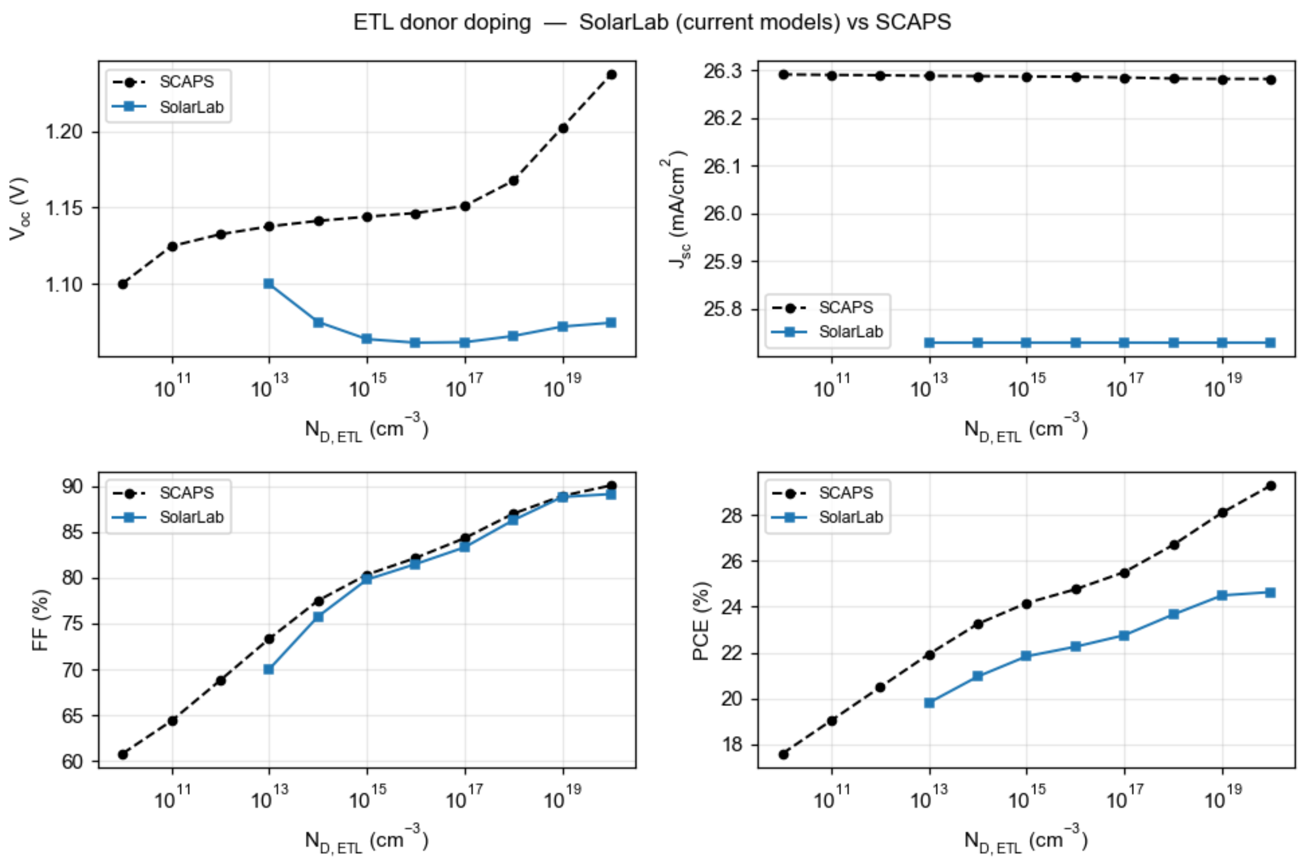

Figure 5: ETL donor doping

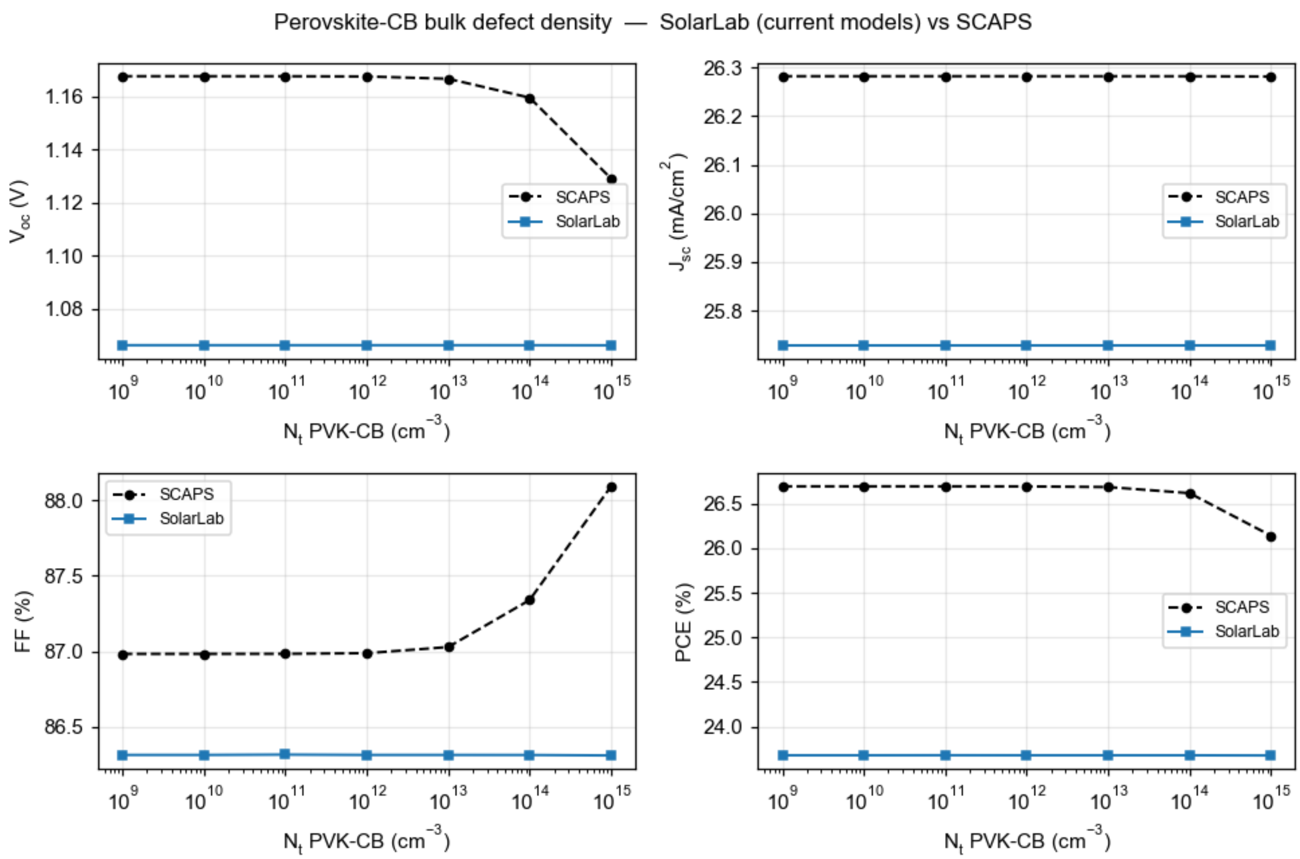

Figure 6: Perovskite-CB bulk defect density

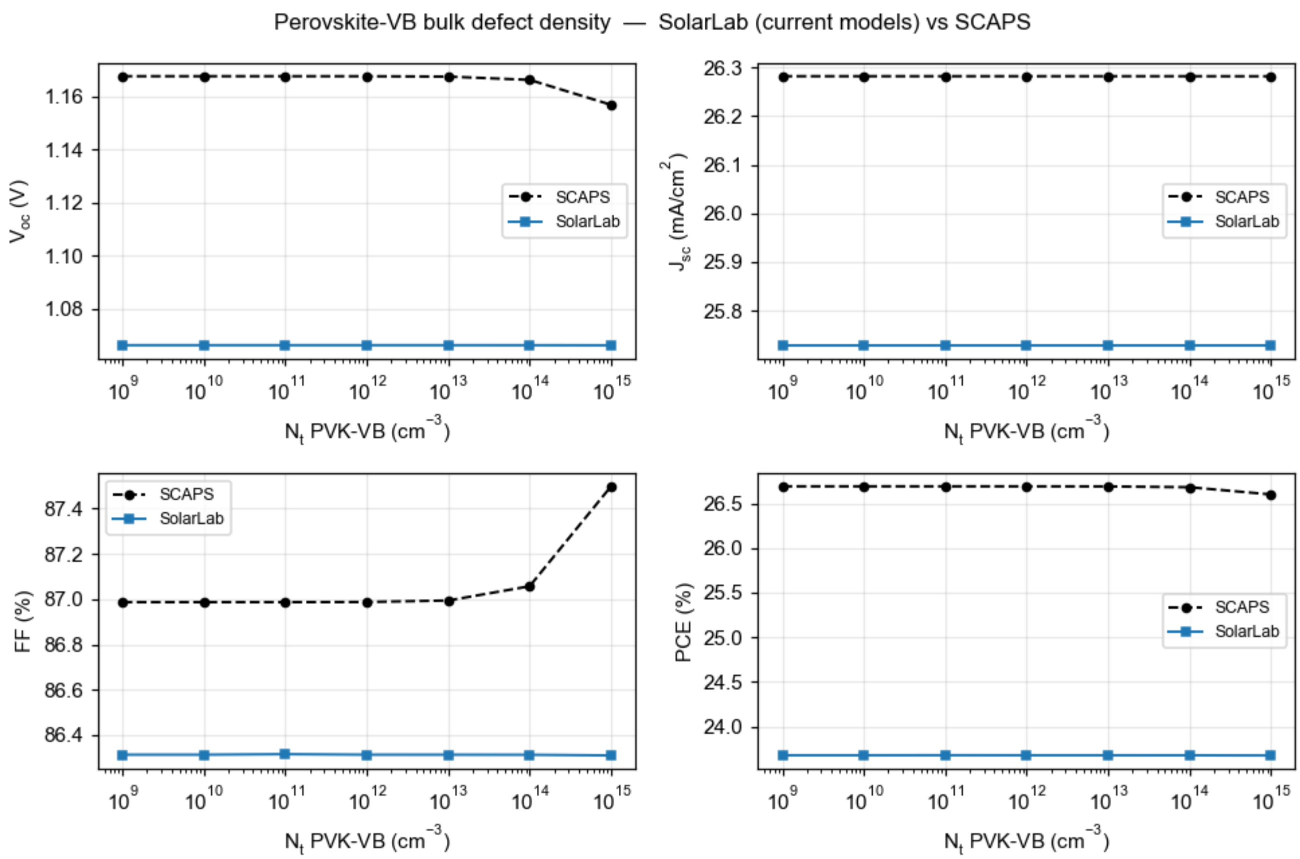

Figure 7: Perovskite-VB bulk defect density

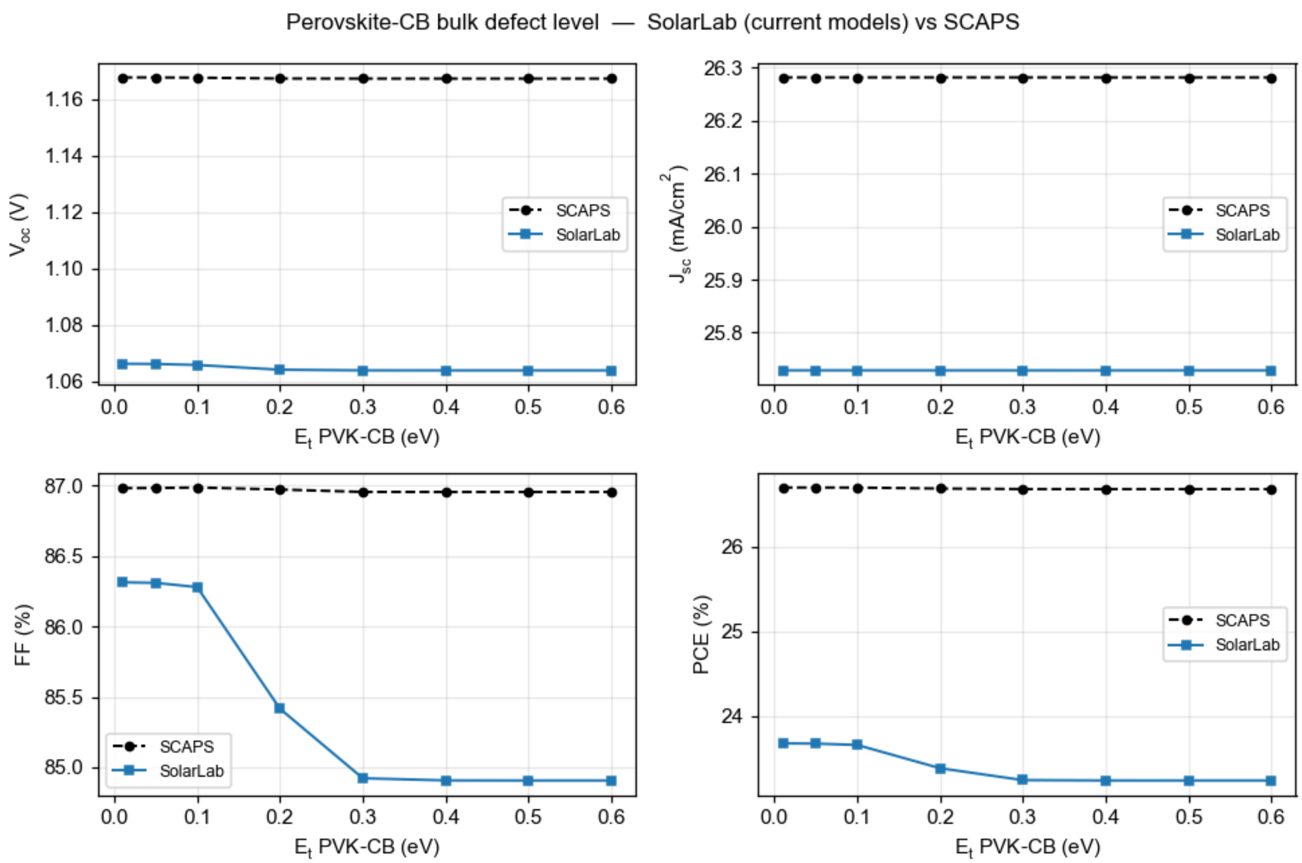

Figure 8: Perovskite-CB bulk defect level

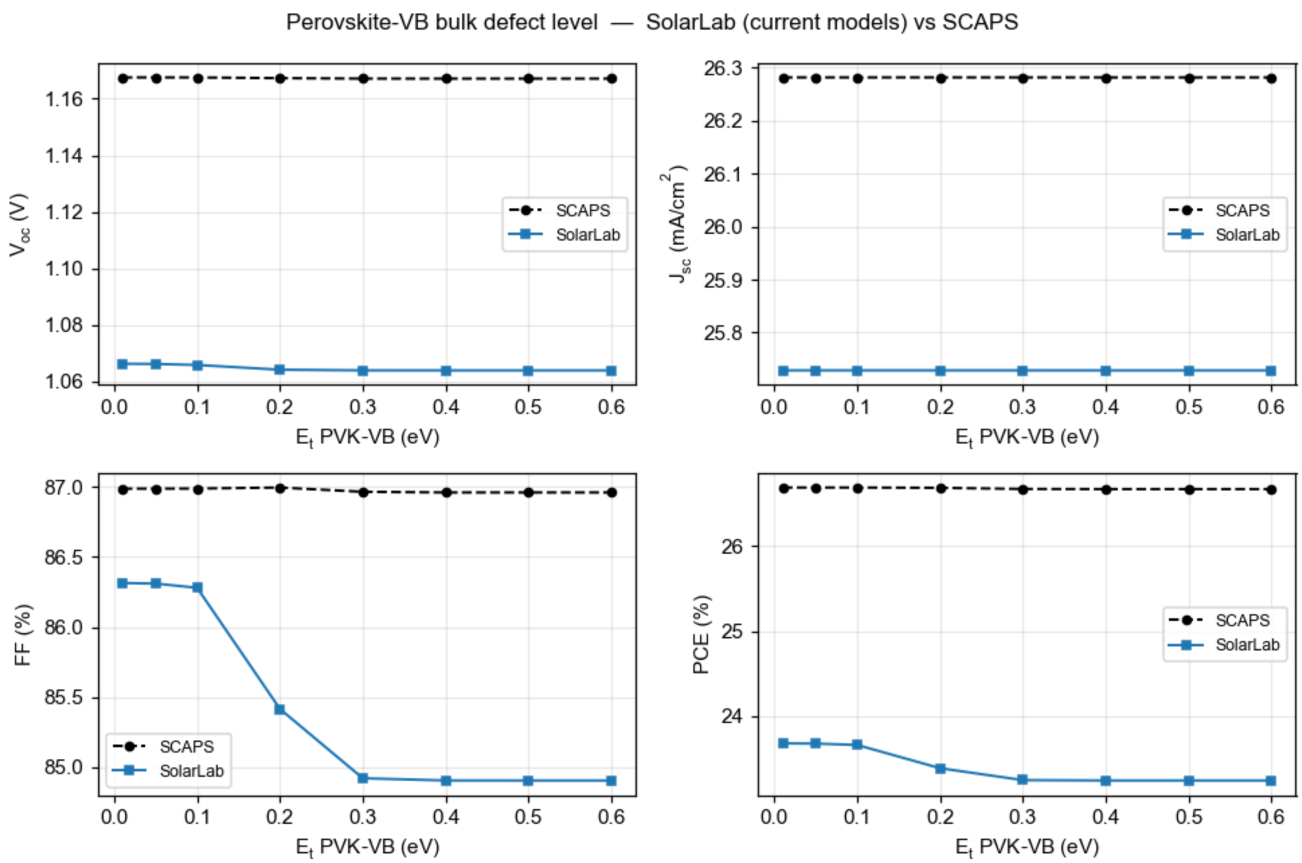

Figure 9: Perovskite-VB bulk defect level

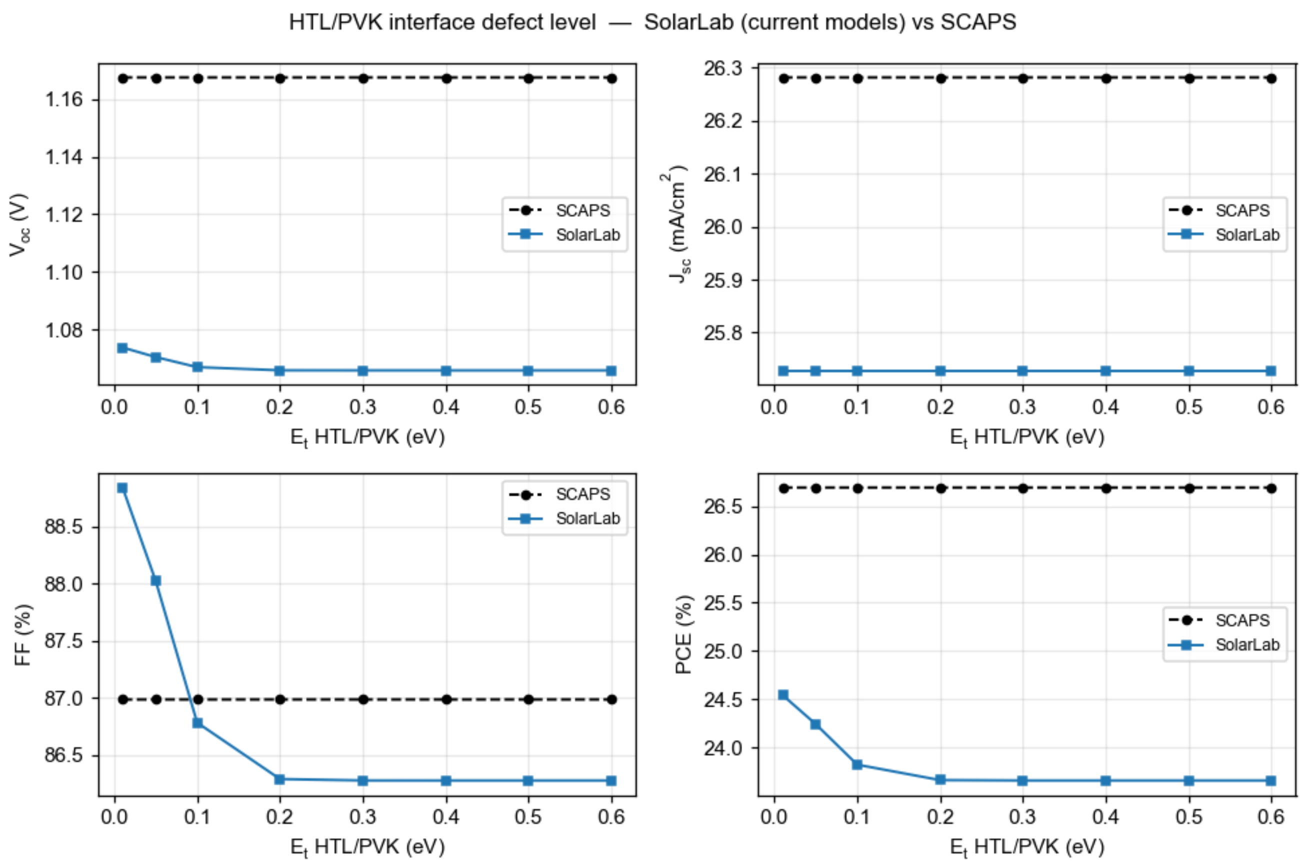

Figure 10: HTL/PVK interface defect level

1. **Base Voc (−96 mV).** With identical ni, recombination coefficients (Auger, radiative) and contact treatment to SCAPS, SolarLab recombines more at a given voltage — the implied dark saturation current is ~37× higher, which is exactly the 93 mV gap (kT·ln 37). At the operating point the recombination is dominated by Auger and radiative channels at the PDF-specified coefficients, plus a ~135 mV internal drop of the quasi-Fermi splitting across the heterojunction band offsets. We verified this is **not** an ni, contact-boundary or interface-calibration error (all tested and ruled out); it is a high-injection carrier-statistics / heterojunction- transport difference between the two solvers.
2. **ETL donor doping (Nd_ETL).** The high-doping arm rises with ND in the correct direction; the low-doping points show a Voc dip tied to the contact / built-in-potential treatment at very low ETL doping, not the interface recombination.
3. **Perovskite bulk Nt (CB / VB).** SolarLab's Voc is pinned by the interface-recombination ceiling, which sits below the voltage where bulk trap density becomes visible; SCAPS reaches that regime and shows a small −39 / −11 mV response. Raising the ceiling requires reducing total recombination, which conflicts with the interface model the matched sweeps rely on.

Jsc's residual −2 % is front-surface reflection that the SCAPS optical model treats as ideal; SolarLab keeps the physical reflection.

Reproducibility

```
cd perovskite-sim
python scripts/scaps_absolute_scorecard.py      # absolute + trend vs the xlsx
python scripts/scaps_validation_figures.py \
       --out ../docs/figures/scaps_validation     # regenerate the overlays
```

Full technical detail (methods, every sweep, and the development history) is in `docs/scaps_validation_report.md`.